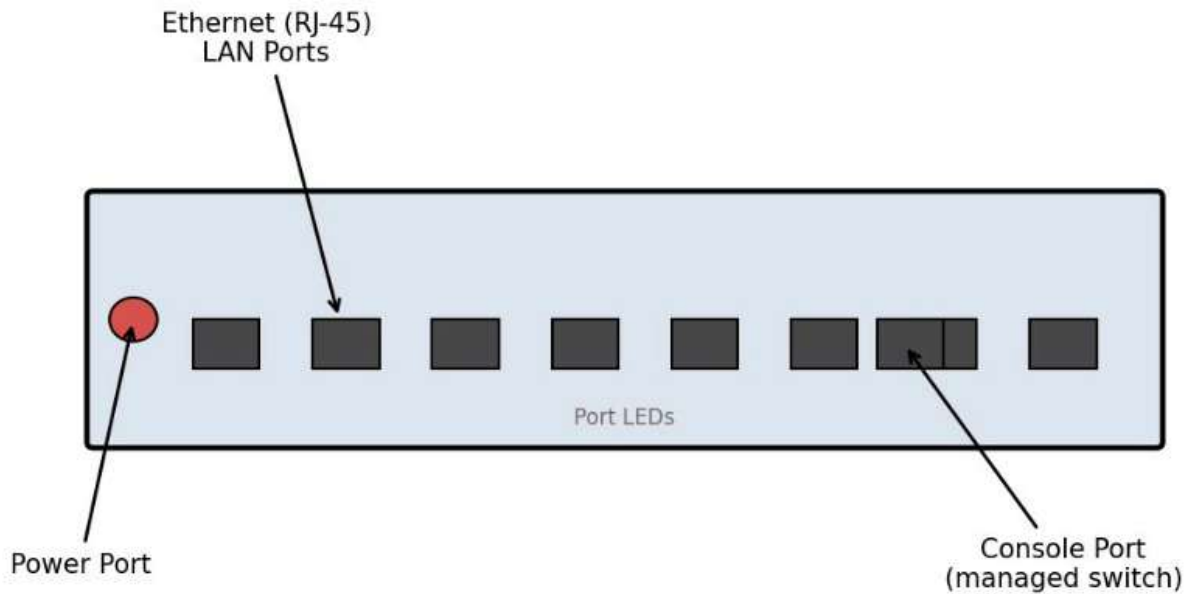


Task 1: Table of main functions and typical ports used by switches, routers, hubs, firewalls, and access points in a home WiFi setup.

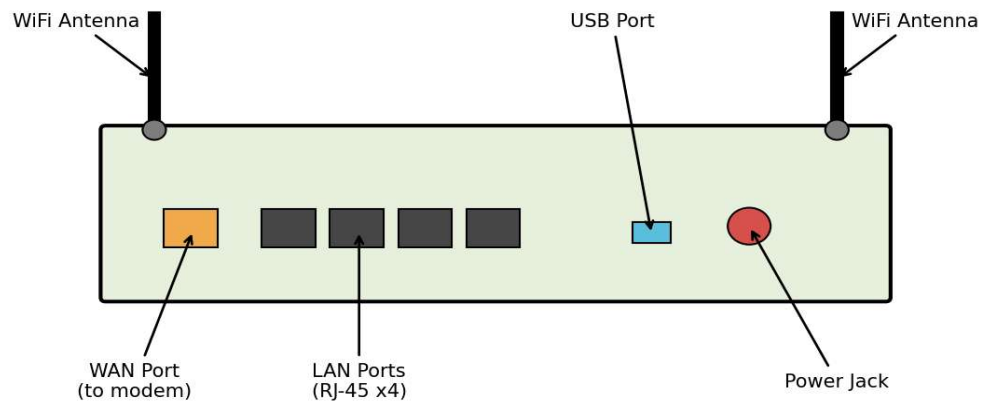
Device	Main Function	Typical Ports
Hub	Broadcasts incoming data to all connected ports (no intelligence — every device shares one collision domain)	4–8 RJ-45 Ethernet ports; power jack
Switch	Connects multiple wired devices on the same network; uses a MAC address table to send data only to the intended device	Multiple RJ-45 Ethernet ports (4–48); console port (RJ-45/USB on managed switches); power jack
Router	Connects different networks (e.g., home LAN to the ISP's internet); handles routing, DHCP, and NAT	1 WAN/Internet port (RJ-45); 4 LAN ports (RJ-45); power jack; sometimes a USB port
Firewall	Filters incoming/outgoing traffic based on security rules; blocks unauthorized access	WAN and LAN Ethernet ports; console port; power jack; dedicated management port
Access Point	Provides WiFi connectivity by bridging wireless devices to the wired network	1 Ethernet uplink port (RJ-45) or PoE port; power jack; antenna connectors (external-antenna models)

Task 2: Labeled diagrams of network devices, marking Ethernet, console, power, and antenna ports.



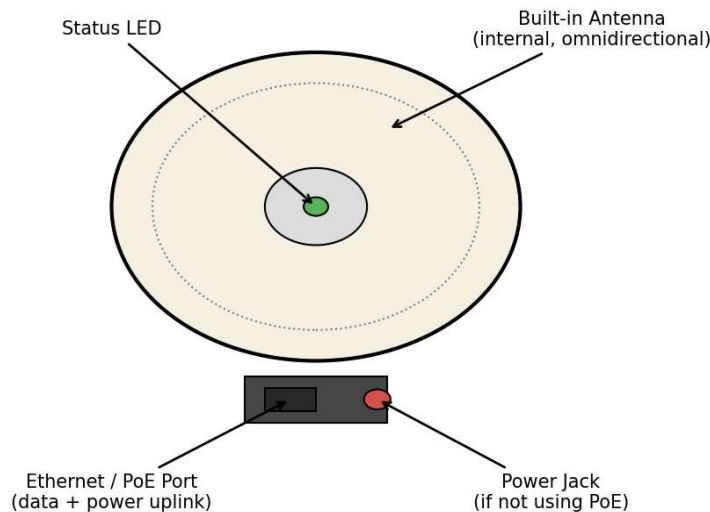
Network Switch — front panel with Ethernet, console, and power ports labeled

Wireless Router — Rear Panel



Wireless Router — rear panel with WAN, LAN, USB, power, and antenna ports labeled

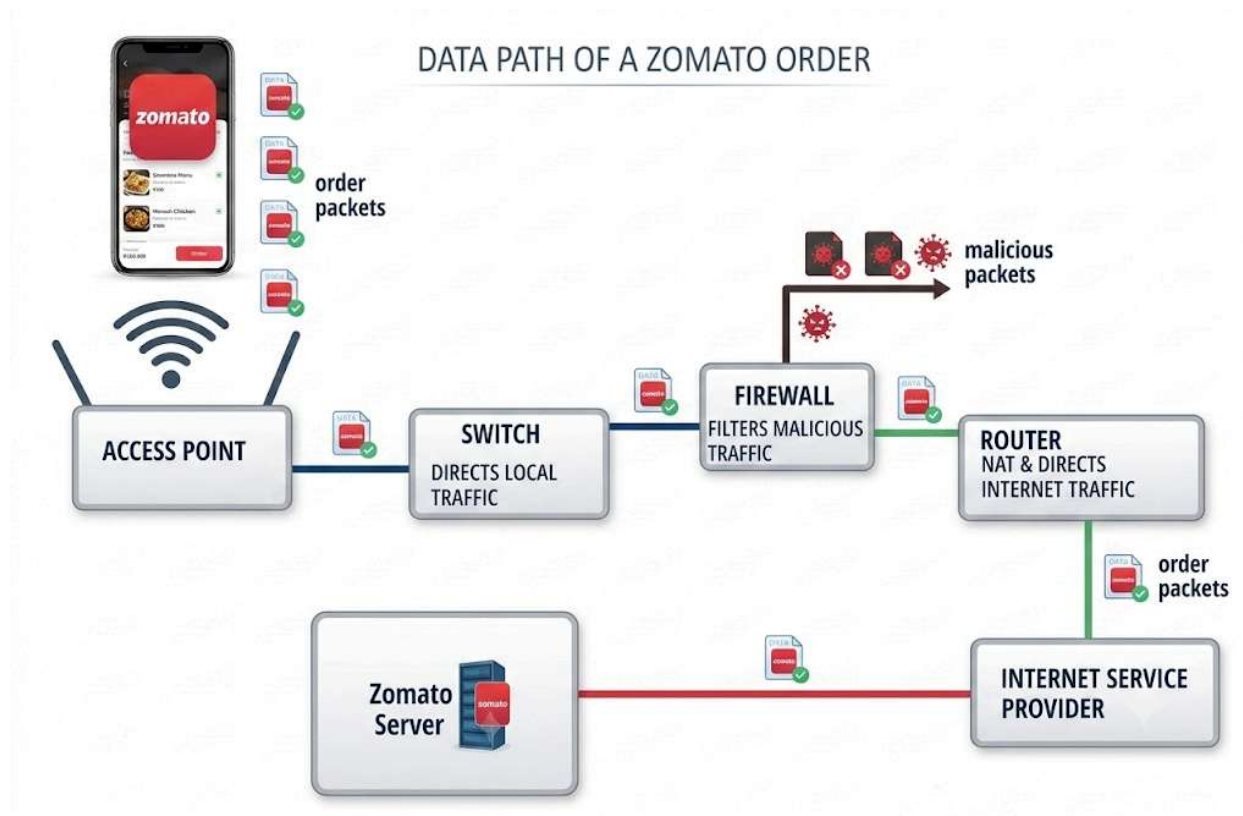
Wireless Access Point (Ceiling-Mount)



Wireless Access Point — showing antenna, status LED, Ethernet/PoE, and power ports

Task 3: Pick a popular app you use daily (like Instagram, Zomato, or Paytm) and describe in 3-4 lines how at least two different network devices (from switches, routers, firewalls, access points) play a role when you use that app.

- **Phone to Wi-Fi:** Your phone sends the request through the air to the **Access Point** (Wi-Fi).
- **Wi-Fi to Wired:** The Access Point hands the request off to the **Switch**, which acts like a smart traffic conductor directing data along wired network cables.
- **Firewall Check (Outbound):** The **Firewall** acts as a security guard, making sure the request going out is safe and permitted.
- **Router & NAT:** The **Router** uses **NAT** (Network Address Translation) to convert your home device's private IP address into a single public IP address so the internet knows where to send data back.
- **ISP to Server:** The request goes across your **ISP Link** (Internet Service Provider) directly to **Zomato's Server**.
- **Return Journey:** Zomato sends the data back. The **Router/NAT** directs it to your phone, and the **Firewall** double-checks the incoming data for safety before it hits your screen.



Task 4: Suppose your college WiFi is slow and sometimes disconnects. Based on the features of access points, switches, and routers, suggest one possible reason for the issue and how upgrading one device could help. Hint: Think about device capacity, coverage, and traffic handling.

Likely reason: College WiFi often relies on a single access point covering a large hall or floor, with far more devices connected than it is built to handle. Each AP supports a limited number of simultaneous connections and shares its total bandwidth across all of them, so during peak hours (e.g., between classes) the AP becomes overloaded, causing slow speeds and dropped connections.

How upgrading helps: Adding more access points or replacing the single AP with a higher-capacity enterprise-grade model, increasing total capacity and covering dead zones. If the bottleneck is instead on the wired side, upgrading the switch connecting the APs to gigabit ports (instead of older 100 Mbps ports) prevents the backbone link from throttling traffic even when the wireless side itself is healthy.